Supplemental Material: Markovian Quantum Neuroevolution for Machine Learning

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In this Supplemental Material, we specifically show how to narrow (enlarge) the gate-block library by imposing (cancelling) some restrictions on building possible gate-blocks. We also mention how to extend the Markovian process in the MQNE algorithm to the high-order Markovian process. Besides, we present more details on the graph-encoding method, the MQNE algorithm, and more numerical results to demonstrate the performance of the proposed scheme.

I. ENCODING VECTORS AND THE PSEUDOCODE

In the main text, we restrict our discussion to the case that the circuits are composed with only single-qubit rotation gates R (composed of $R_z R_x R_z$, where R_x and R_z denote rotations along x and z axes, respectively) and two-qubit controlled-R_x gates, and the controlled-R_x gates only act on adjacent qubits. The pseudocode for the MQNE algorithm is provided in Algorithm S1.

II. THE GATE-BLOCK LIBRARY

For a k-qubit quantum gate-block library, we can sort all quantum gate-blocks into the ones containing 0 controlled-R_x gate, 1 controlled-R_x gate, until $\lfloor k/2 \rfloor$ controlled-R_x gates.

For quantum gate-blocks containing 0 controlled-R_x gate, each of k qubits can be acted on by a single-qubit rotation gate R or an identity gate, corresponding to 2^k quantum gate-blocks of this type totally. Likewise, for quantum gate-blocks containing 1 controlled-R_x gate, each of the remaining $k-2$ qubits can be acted on by rotation gate R or identity gate pertaining to 2^{k-2} possibilities. Through some combinatorial calculations, we can obtain the total number of 1 controlled-R_x blocks:

$$N_1(1) = 2^{k-2} \times \binom{k-1}{1} \times 2^1 .$$

Similarly, for quantum gate-blocks containing i controlled-R_x gate, where $i \leq \lfloor k/2 \rfloor$, each of the remaining $k-2i$ qubits can be acted on by gate R or the identity gate, we obtain

$$N_1(i) = 2^{k-2i} \times \binom{k-i}{i} \times 2^i .$$

Then the total number of quantum gate-blocks is:

$$f_1(k) = \sum_{i=0}^{\lfloor k/2 \rfloor} N_1(i) = \frac{(1+\sqrt{3})^{k+1} - (1-\sqrt{3})^{k+1}}{2\sqrt{3}} .$$

In this way, we have calculated there are $f_1(k) = \Theta[(1+\sqrt{3})^k]$ quantum gate-blocks in the gate-block library used in the main text. To analyse the complexity of the searching space in the MQNE algorithm, we also numerically study how the outdegree (for a node, its outdegree is the number of other nodes it is allowed to point to) of nodes scale with respect to qubit number and show numerical results in Fig. S2, from which it is evident that the maximum (average) value of

Algorithm S1 Markovian quantum neuroevolution algorithm

Input: The directed graph, hyperparameters n_i, t_i, l, l', f_c, and g_c
Output: The optimal quantum circuit architecture
Initialization: start from a fixed node and uniformly sample n_1 paths with length-l, and compute their fitness
for $i = 1$ to g_c **do**
  Choose the t_i ($t_i < n_i$) paths with highest fitness
  Evolution: produce paths in the $(i+1)$-th generation by concatenating n_{i+1} segments of length l' to each of the t_i paths, and compute their fitness
  if max [fitness (paths)] $\geq f_c$ **then**
    Terminate the iteration
  end if
end for
Output the optimal quantum circuit

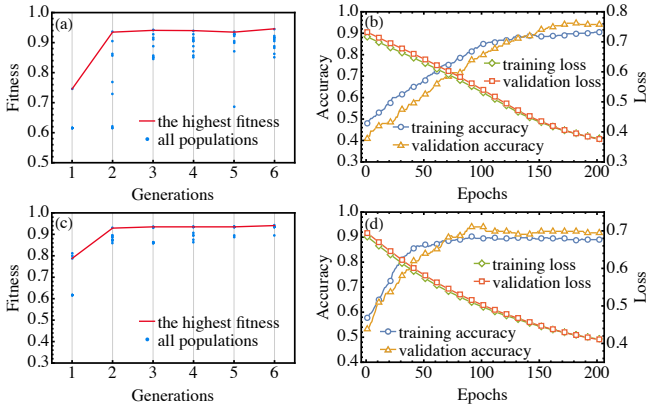


FIG. S1. The numerical results for applying the MQNE algorithm to the task of classifying samples in the cancer dataset. In (a), we randomly choose the initial variational parameters of single-qubit rotation gates at the beginning of the training process. In (b), the loss and accuracy are plotted as functions of training epochs for the sixth-generation quantum circuit with the highest fitness. (c) and (d) are similar to (a) and (b), but with fixed initial variation parameters.

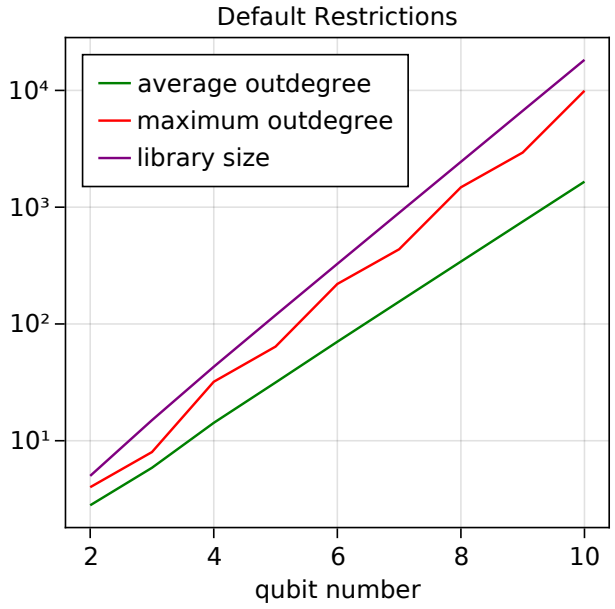


FIG. S2. The size of the gate-block library, and the maximum (average) outdegree of all nodes in the directed graph as functions of qubit number under default restrictions in the main text.

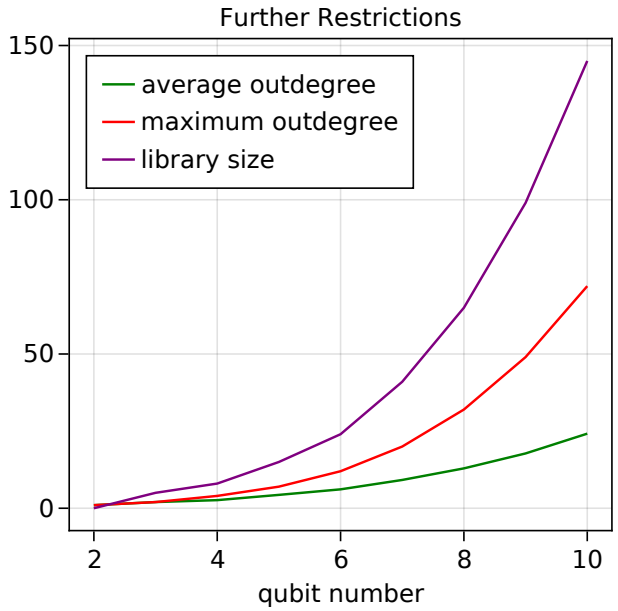


FIG. S3. We plot the size of the gate-block library, and the maximum (average) outdegree of all nodes in the directed graph as functions of qubit number under further restrictions we added.

the outdegree has the same growing trend as the library size, which will climb exponentially as qubit number increases.

In some realistic problems, considering limited classical computational power or other realistic reasons, especially for large qubit number case, it is necessary to impose further restrictions to reduce the size of the library, from the exponential scaling $f_1(k)$ to a polynomial scaling $f_2(k)$. in our proposal, We plus three more restrictions. Firstly, for quantum gate-blocks containing i controlled-R_x gate, we assume the remaining $k-2i$ qubits are acted on *simultaneously* by either gate R or the identity gates. Secondly, for each controlled-R_x gate acting on two neighbouring qubits i and $i+1$, we assume that the i-th qubit is the controlling qubit. Thirdly, we restrict each quantum gate-block contains at most c (a cut-off constant) controlled-R_x gates.

With these further restrictions, there are $N_2(i) = \binom{k-i}{i}$ gate-blocks containing i controlled-R_x gates, and consequently the total number of gate-blocks $f_2(k)$ in the library is given by

$$f_2(k) = \sum_{i=0}^{c} N_2(i) = \sum_{i=0}^{c} \binom{k-i}{i} = O(k^c).$$

And we also numerically explore how the outdegree scale with respect to qubit number with numerical results shown in Fig. S3, which reflects that the maximum (average) value of the outdegree which will rise polynomially with respect to qubit number.

Similarly, adding more restrictions can even reduce the size of the library to a constant f_3, independent of k. For instance, a gate-block library containing only three gate-blocks can be constructed to obtained a quantum classifier architecture as shown in Fig. S4.

We mention that, the gate-block marked yellow corresponds to the rotation layer, the gate-blocks marked orange and green correspond to the entanglement layer respectively, in the Fig.2 of Ref. [1].

On the contrary, in order to improve the possibility of finding the optimal quantum circuits, we can enlarge the searching space by releasing some restrictions mentioned in the main text and above. For example, we may remove the restriction that the controlled-R_x gates only act on adjacent qubits. We obtain the total number of quantum gate-blocks containing 1 controlled-R_x gate:

$$N_0(1) = 2^{k-2} \times \mathrm{A}_k^2 \,.$$

Likewise, for quantum gate-blocks containing i controlled-R_x gates, where $i \leq \lfloor k/2 \rfloor$, with each of the remaining $k-2i$ qubits being acted on by gate R or identity gate, we obtain

$$N_0(i) = 2^{k-2i} \times \mathrm{A}_k^{2i} / \mathrm{A}_i^i \,.$$

Then the total number of gate-blocks is

$$f_0(k) = \sum_{i=0}^{\lfloor k/2 \rfloor} N_0(i) \,.$$

In this way, the size of the quantum gate-block library is enlarged to $\Omega(\lfloor k/2 \rfloor !)$.

In practical applications, we can directly construct the gate-block library for $(k+1)$-qubit circuits based on existed libraries for $(k-1)$- and k-qubit circuits. Firstly, the operations on the top k qubits can be adopted entirely from the gate-block

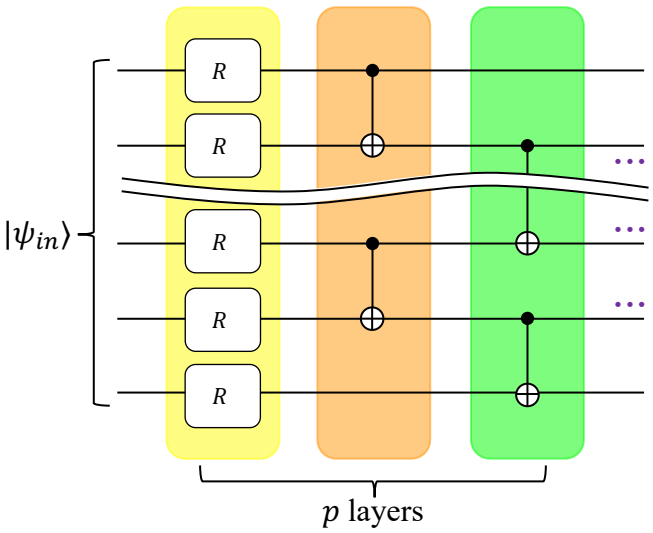


FIG. S4. The sketch of a quantum classifier architecture. It consists of p same layers, each layer has three gate-blocks, the gate-block marked yellow is the rotation unit, the gate-blocks marked orange and green are in the entanglement unit.

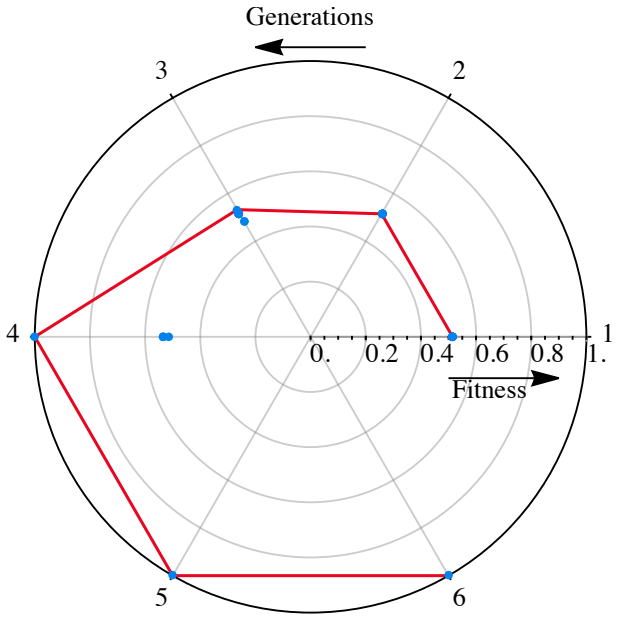


FIG. S5. The performance of the MQNE algorithm in the task of classifying symmetry-protected topological states, with the initial variational parameters fixed at the beginning of the training process.

library of k-qubit circuits when the last qubit is acted on by gate R or identity gate. Secondly, the operations on the top $(k-1)$ qubits can be adopted entirely from the gate-block library of $(k-1)$-qubit circuits when the last two qubits is acted on by a controlled-R_x gate. In this way, we can readily scale up the gate-block library of $(k-1)$-qubit and k-qubit circuits to the gate-block library of $(k+1)$-qubit circuits.

III. THE MQNE ALGORITHM WITH HIGH-ORDER MARKOVIAN PROCESS

In the main text, we consider the rules between adjacent gate-blocks with the help of the single-qubit rotation gate R. We could regard finding paths in the directed graph as a Markovian process, in which the future state depends only on the current state of the system, but not on the previous ones. All paths of the graph correspond to the discrete state space, and the adjacency matrix of the graph corresponds to the transition matrix describing the probabilities of transitions between states. In the main text, we consider a simple case where the transition probability is uniformly distributed on all states that the present state can transit to.

For the high-order Markovian process, we could directly use R_z and R_x gates, instead of the composed R gate, to construct quantum gate-blocks. Consequently, we need to invoke new rules between adjacent four gate-blocks to avoid ambiguity and duplication of successive rotations. For example, when three consecutive quantum gate-blocks contain R_z, R_x, R_z on the same qubit respectively, then the fourth gate-block should not have R_z or R_x on this qubit. So the future state depends on the previous three states of the system, which is a third-order Markovian process.

IV. NUMERICAL RESULTS FOR THE CANCER DATASET

Another example we consider is the classification of samples in the cancer dataset [2], we need six qubits to encode each input sample and use an additional qubit to output the results of the binary classification (whether the label is cancer or not). Thus, the ansatz circuit we aim to design is a seven-qubit variational circuit. Applying the graph-encoding method for seven-qubit circuits and supposing that controlled-R_x gates act only on adjacent qubits, we obtain 896 gate-blocks and the corresponding directed graph has 896 nodes. Based on the connection rules, we compute the adjacency matrix and apply the MQNE algorithm with hyperparameters set as $(n_i, t_i, l, l') = (10, 1, 3, 2)$.

Our results are summarized in Fig. S1. In Fig. S1(a), we randomly choose the initial variational parameters for single-qubit rotation gates when training the generated quantum classifiers at each generation. The MQNE algorithm outputs a quantum circuit (with 68 variational parameters) with fitness 94.6% at the sixth generation. The corresponding path for this circuit on the directed graph reads $[895 \to 31 \to 801 \to 273 \to 404 \to 643 \to 423 \to 593 \to 225 \to 481 \to 673 \to 792 \to 609]$, where the numbers denotes the labels of the nodes of the graph.

In Fig. S1(b), we plot the average accuracy and loss for both the training and validation datasets as a function of the number of epochs during the training process. After training, the performance of this quantum classifier is also tested on the testing dataset and an accuracy of 94.6% is obtained. We mention that the numerical simulations of training the quantum classifiers are based on the Julia language [3] and Yao.jl

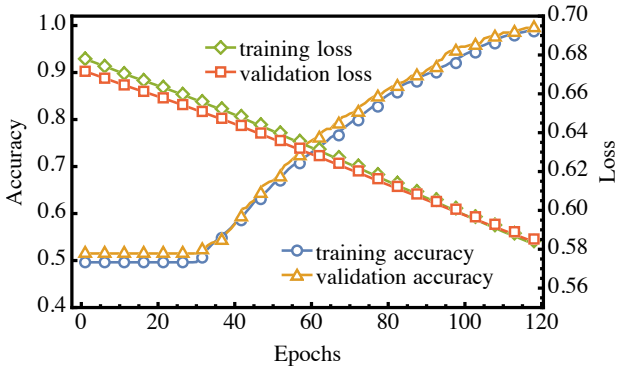


FIG. S6. The numerical results for classifying symmetry-protected topological states of the cluster-Ising model. The initial variational parameters of single-qubit rotation gates are randomly chosen during the training process, we plot the loss and accuracy as a function of training epochs for the sixth-generation quantum circuit with the highest fitness.

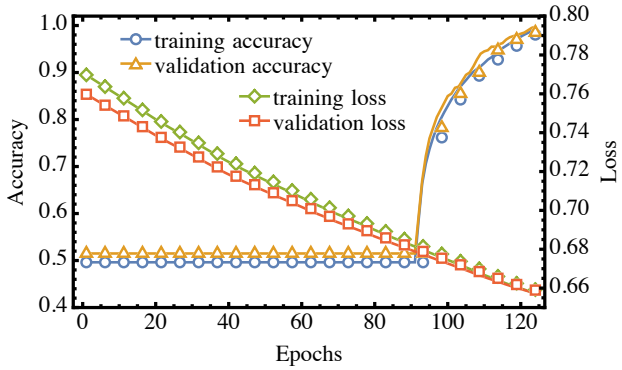


FIG. S7. The loss and accuracy as a function of training epochs for the sixth-generation circuit with the highest fitness (see Fig. S5), for the task of classifying symmetry-protected topological states of the cluster-Ising model.

[4] framework, throughout the paper. Fig. S1 (c) and (d) are analogous to Fig. S1 (a) and (b) respectively, but with fixed initial parameters during the training process.

V. MORE NUMERICAL RESULTS FOR CLASSIFYING HANDWRITTEN-DIGITS AND SPT STATES

The MNIST dataset consists of two-dimensional gray-scale images (28×28 pixels) for handwritten digits from 0 through 9, here we display four gray-scale images for digits 1, 4, 7 and 9 from MNIST dataset in Fig. S10.

In the classification of handwritten-digit images in the MNIST dataset, the MQNE algorithm outputs a quantum circuit with fitness (accuracy) 97% at the sixth generation, whose structure is explicitly shown in Fig. S11.

In the main text, we also consider a quantum machine learning task of classifying SPT states. We use two strategies to train the generated quantum classifiers at each generation. Firstly, we randomly choose the initial variational parameters for single-qubit rotation gates when training the generated quantum classifiers at each generation, our results are plotted in the main text. The MQNE algorithm outputs a quantum circuit (with 46 variational parameters) with fitness 100% at the third generation. The corresponding path for this circuit on the directed graph reads $[6687 \rightarrow 1665 \rightarrow 2177 \rightarrow 2401 \rightarrow 4899 \rightarrow 3042 \rightarrow 4901]$. In Fig. S6, we plot the average accuracy and loss for both the training and validation datasets as a function of the number of epochs during the training process.

Secondly, we fix initial parameters for single-qubit rotation gates during the training process, the results are shown in Fig. S5. The MQNE algorithm outputs a quantum circuit (bearing 54 variational parameters) with fitness 100% at the fourth generation. The corresponding path for this circuit on the directed graph reads $[6687 \rightarrow 3649 \rightarrow 3173 \rightarrow 737 \rightarrow 1798 \rightarrow 60 \rightarrow 1217 \rightarrow 4905 \rightarrow 1826]$, In Fig. S7, we plot the average accuracy and loss for both the training and validation datasets as a function of the number of epochs during the training process.

VI. PARAMETER SETTINGS IN TRAINING QUANTUM CLASSIFIERS

In the task for classification of handwritten-digit images in the MNIST dataset (see Fig. 2 in the main text), we use the Adam optimizer with a batch size of 30 and a learning rate of 0.0015 to minimize the loss function. The training step is 200. The fitness is averaged on 2000 training samples and 500 validation samples which are not contained in the training dataset.

In the task for classification of samples in the cancer dataset (see Fig. S1), we use the Adam optimizer with a batch size of 20 and a learning rate of 0.0015 to minimize the loss function. The training step is 150. The fitness is averaged on 400 training samples and 169 validation samples which are not contained in the training dataset.

In the task for classifying symmetry-protected topological states (see Fig. 3 in the main text, and Figs. S5, S7 and S6 in this Supplemental Material), we use the Adam optimizer with a batch size of 20 and a learning rate of 0.0015 to minimize the loss function. The training step is 150. The fitness is averaged on 1600 training samples and 400 validation samples which are not contained in the training dataset.

VII. THE DIRECT GENERALIZATION OF THE NEAT ALGORITHM IN SEARCHING QUANTUM CIRCUITS

In this part, we show the performance of the genetic algorithm, which could be regarded as a naive generalization of the NEAT algorithm [5], in searching quantum circuits.

In conventional evolutionary algorithms, we evolve the population to create the next generation by applying genetic

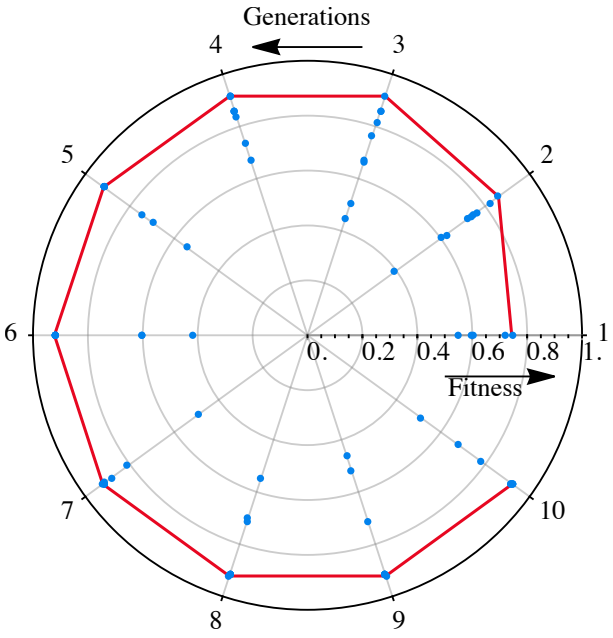


FIG. S8. The performance of the genetic algorithm in the task of classifying handwritten-digit images in the MNIST dataset.

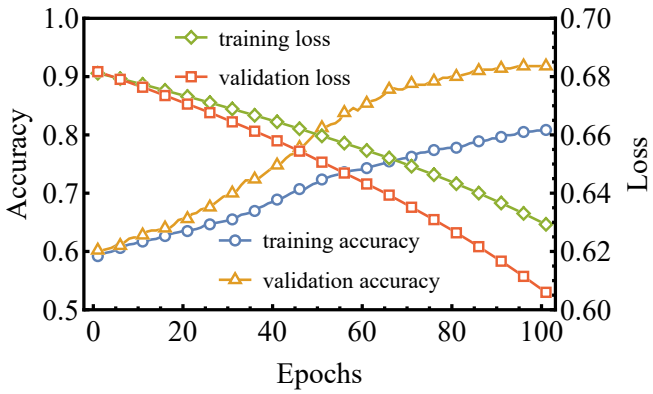


FIG. S9. The loss and accuracy as a function of training epochs for the tenth-generation quantum classifier with the highest fitness, in classifying handwritten-digit images in the MNIST dataset. Here, we use the genetic algorithm to construct quantum circuits.

operators on individuals to generate offspring. Two important genetic operators are mutation and crossover [5, 6], where crossover produce new offspring by recombining two selected individuals (parents), and mutation generate new offspring by randomly mutating a selected individual. Applying them in searching quantum circuits, crossover is defined to divide each of two parent quantum circuits into two parts respectively and exchange their divided parts to create two offspring, mutation is defined to randomly generate some positions in circuits and then replace quantum gates in these positions with other different gates.

We design the genetic algorithm as shown in Algorithm S2 to search for optimal quantum classifier architectures (9-qubit circuits) for the MNIST handwritten digit dataset. The major hyperparameters (n_i, t_i) are set as $(9, 3)$.

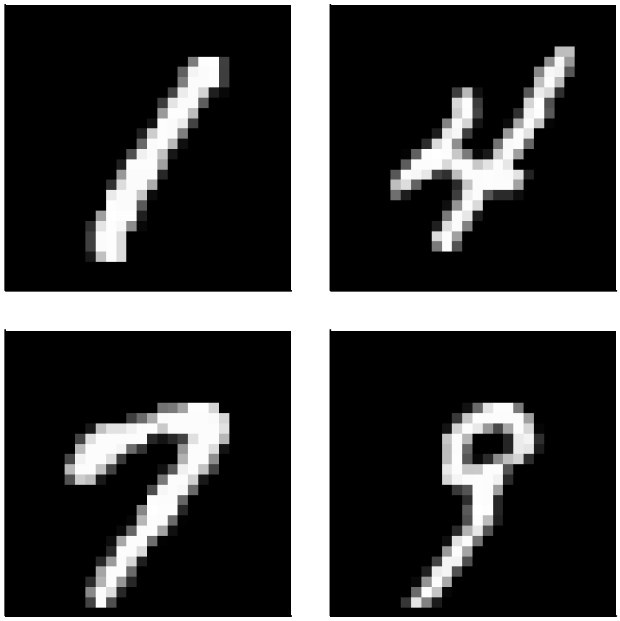


FIG. S10. Images for handwritten digits from MNIST dataset: 1, 4, 7, 9 [7].

Fig. S8 displays the fitness of 86 quantum circuits totally evaluated in running the genetic algorithm. We find a circuit structure with fitness 92% at the third generation, whose number of quantum gates is more than 100. In Fig. S9, we plot the average accuracy and loss for both the training and validation datasets as a function of the number of epochs during the training process. We see from Fig. S8 that the local convergence appears at the third generation and the fitness does not increase for later generations. The performance of the genetic algorithm is ineffective when directly applying these two operators in designing quantum classifiers.

Algorithm S2 The genetic algorithm

Input: Hyperparameters n_i, t_i, g_c, f_c, etc
Output: The optimal quantum circuit architecture
Initialization:: randomly generate n_1 quantum circuits, and compute their fitness
for $i = 1$ to g_c **do**
  Choose the best t_i ($t_i < n_i$) circuits with highest fitness
  Apply the crossover and mutation operator
  Compute the fitness of individuals in the $(i+1)$-th generation
  if max[fitness(circuits)] $\geq f_c$ **then**
    Terminate the iteration
  end if
end for

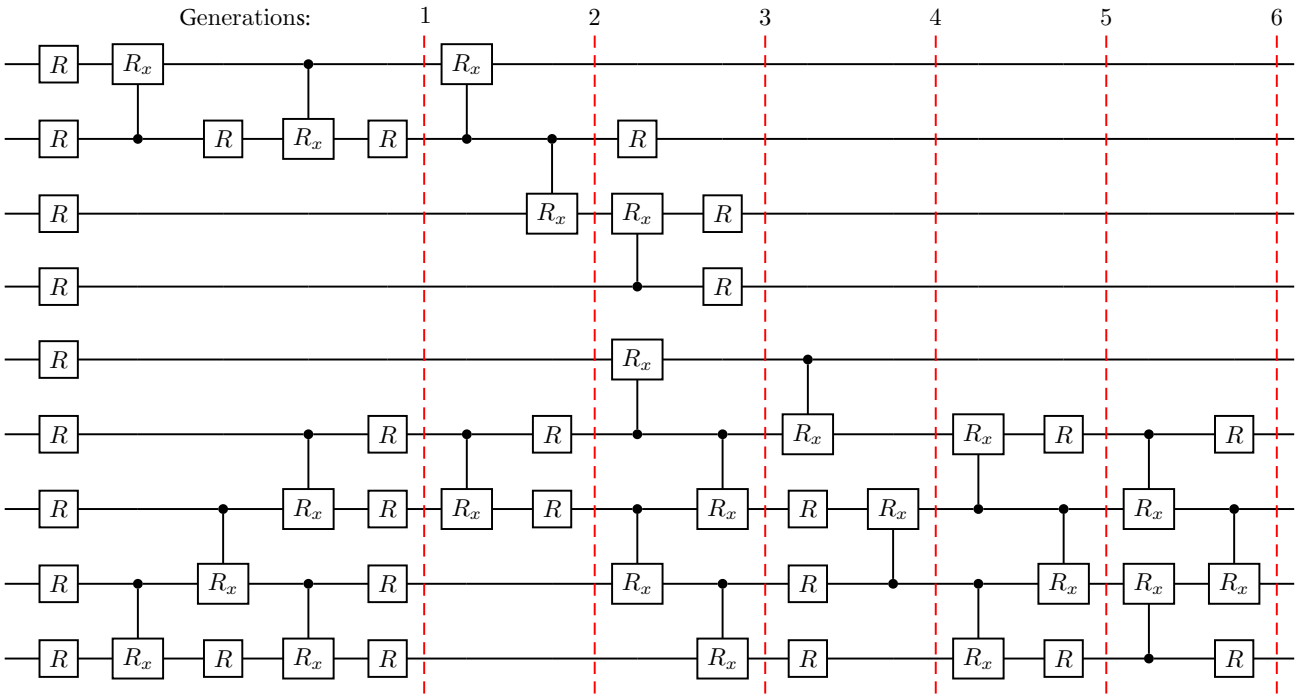


FIG. S11. The optimal structure of the quantum circuit generated by the MQNE algorithm to classify handwritten-digit images in the MNIST dataset. The numbers in the top denote the generations.

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